MAHARASHTRA CYBER FORENSIC DEPARTMENT

DIGITAL FORENSICS & INCIDENT RESPONSE (DFIR) UNIT — STATE OF MAHARASHTRA

OFFICIAL CYBER FORENSIC INVESTIGATION REPORT

INCIDENT ASSESSMENT: RANSOMWARE / EXTORTION

RESTRICTED — MAHARASHTRA LAW ENFORCEMENT / SOC CONFIDENTIAL — CHAIN OF CUSTODY PRESERVED

Case Identifier:	CASE-MH-2026-0001	**Report Version:**	v2.5 Enterprise DFIR
Target Evidence:	`ransomware_incident_syslog.log`	**Report Date:**	23-Oct-2026 13:41:24 UTC+05:30
Incident Severity:	**CRITICAL (100/100)**	**Investigation Status:**	Active Containment / Triage Complete
Primary Analysts:	Pragati Patil, Tanisha Lohar, Gaurav Rajguru	**Project Guide:**	Ms. Tejas P. Shinde
Department:	Department of Computer Science and Engineering (IoT & Cyber Security including Blockchain)	**Institution:**	ADCET, Ashta, Maharashtra (Autonomous)

1. EXECUTIVE SUMMARY

The Maharashtra Cyber Forensic Department (DFIR Unit) initiated an automated digital forensic triage of the evidence artifact **'ransomware_incident_syslog.log'** (2,646 bytes) to ascertain unauthorized activity, malicious payload execution, lateral movement, and data exfiltration. The automated triage engine classified the incident with an overall risk severity rating of **CRITICAL (Risk Score: 100/100)**.

Supervised Machine Learning classification (Random Forest) identified the primary threat vector as **'Ransomware / Extortion'** with **63.0% confidence**. The attack has advanced to Cyber Kill Chain stage: **'Critical Stage - Active Threat Impact / Ransomware Execution'** (100.0% progression). Unsupervised Anomaly Modeling (Isolation Forest) confirmed system deviation status as **'Anomalous'** (Anomaly Index: **0.00**).

Analysis discovered a cumulative total of **6 Indicators of Compromise (IOCs)**, **3 heuristic YARA signature matches**, and **5 correlated MITRE ATT&CK; techniques**. Immediate endpoint containment, credential revocation, and perimeter firewall blocking are required.

2. SCOPE AND METHODOLOGY

The scope of this forensic triage encompasses the examination of digital evidence submitted to the Cyber Triage platform. The analysis followed a forensically sound methodology compliant with ISO/IEC 27037 standards for digital evidence handling:

Evidence Artifact	SHA-256 Cryptographic Hash	Size	Format / Status
`ransomware_incident_syslog.log`	`dd0be9cf2a87b54bed5cfee78b9f9dede56aa0b2419c69116f69fb1052a2f57d`	2,646 B	LOG (Preserved)

- **Cryptographic Verification:** Bit-level hashing (SHA-256, MD5, SHA-1) upon intake to ensure zero evidence tampering.
- **Pattern & IOC Extraction:** Automated extraction of IPv4/IPv6, URLs, domains, hashes, CVEs, and Base64 payloads.
- **YARA Heuristic Scanning:** Heuristic detection of ransomware strings, shadow copy deletion, Mimikatz, and webshells.
- **Machine Learning Modeling:** Dual AI architecture combining Isolation Forest anomaly scoring with Random Forest classification.
- **Threat Intelligence Enrichment:** Live reputation correlation against known threat actor infrastructures and C2 gateways.
- **ATT&CK; & Kill Chain Mapping:** Normalization of forensic artifacts into MITRE Enterprise Tactics and Kill Chain stages.

3. AI ANOMALY & THREAT CLASSIFICATION RESULTS

Analytical Metric	Quantitative Value	Forensic Interpretation
Overall Threat Risk Score	**100 / 100**	Calibrated severity level: CRITICAL
Supervised Threat Classifier (RF)	**Ransomware / Extortion**	Multi-class probability distribution: Ransomware / Extortion: 63.0% | Credential Theft & Access: 36.1% | Web Application Compromise: 1.0%
Isolation Forest Anomaly Index	**0.00 / 1.00**	Unsupervised deviation status: Anomalous
Cyber Kill Chain Progression	**100.0%**	Lifecycle stage reached: Critical Stage - Active Threat Impact / Ransomware Execution
Shannon Binary Entropy	**5.54 / 8.00**	Measures code randomness, packing, or encryption density

4. MITRE ATT&CK; ENTERPRISE MAPPING

Tactic Name	Technique ID	Technique Name	Observed Artifact / Trigger
Impact	**T1490**	Inhibit System Recovery	`vssadmin`
Impact	**T1486**	Data Encrypted for Impact	`lockbit`
Execution	**T1059.001**	Command and Scripting Interpreter: PowerShell	`powershell`
Initial Access	**T1190**	Exploit Public-Facing Application	`rce`
Defense Evasion	**T1070.001**	Indicator Removal: Clear Windows Event Logs	`wevtutil`

5. INDICATORS OF COMPROMISE (IOCs)

5.1 Network Indicators (Defanged)

Type	Defanged Value	Threat Context / Reputation	Confidence
IP Address	`185[.]220[.]101[.]5`	Malicious C2 / TOR Exit Node | LockBit 3.0 / APT29	95%
IP Address	`194[.]26[.]29[.]112`	Active Ransomware C2 Gateway | LockBit Affiliate	90%
URL / Domain	`hxxp://185[.]220[.]101[.]5/payload/stage2[.]e`	Outbound HTTP Request	90%

5.2 Host & Endpoint Indicators

Type	Indicator / Artifact Path	Severity & Forensic Context
SHA-256	`e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855`	Cryptographic Hash / Malware Sample
MD5	`7b2c9a6234b071e62a8291a13426e95c`	Cryptographic Hash / Malware Sample
LOLBAS Command	`powershell.exe -nop -w hidden -enc`	Severity: High (Living off the Land)
LOLBAS Command	`vssadmin.exe delete shadows`	Severity: Critical (Living off the Land)
LOLBAS Command	`wmic shadowcopy delete`	Severity: High (Living off the Land)

6. CHRONOLOGICAL INCIDENT TIMELINE

Timestamp (UTC)	Event Description	Category	Severity
`2026-08-17 02:14:22`	Aug 17 02:14:22 DC01-SRV Microsoft-Windows-Security-Aud	System Event	**Info**
`2026-08-17 02:14:45`	Aug 17 02:14:45 DC01-SRV Microsoft-Windows-PowerShell[4	Suspicious Execution / LOLBAS	**High**

2026-08-17 02:15:10	Aug 17 02:15:10 DC01-SRV Microsoft-Windows-Sysmon[1]: P	Ransomware / Critical Threat	Critical
2026-08-17 02:15:33	Aug 17 02:15:33 DC01-SRV Microsoft-Windows-Sysmon[1]: P	Ransomware / Critical Threat	Critical
2026-08-17 02:16:01	Aug 17 02:16:01 DC01-SRV Microsoft-Windows-Sysmon[1]: P	Ransomware / Critical Threat	Critical
2026-08-17 02:16:08	Aug 17 02:16:08 DC01-SRV Microsoft-Windows-Sysmon[1]: P	System Event	Info
2026-08-17 02:16:15	Aug 17 02:16:15 DC01-SRV Microsoft-Windows-Sysmon[1]: P	System Event	Info
2026-08-17 02:16:22	Aug 17 02:16:22 DC01-SRV Microsoft-Windows-Sysmon[1]: P	System Event	Info

7. YARA & HEURISTIC DETECTION SUMMARY

• **[RULE-RANSOM-001] Ransomware Shadow Copy & Backup Invalidation** (Critical Severity)

Details: Attempts to inhibit system recovery by deleting volume shadow copies or disabling startup repair.

• **[RULE-RANSOM-002] Ransomware Note & Known Extortion Signatures** (Critical Severity)

Details: Indicators of ransom notes, encryption extensions, or extortion group signatures (e.g. LockBit, Conti).

• **[RULE-EVASION-001] Defense Evasion & Security Log Tampering** (High Severity)

Details: Attempts to clear event logs, disable antivirus, or bypass execution policies.

8. CONTAINMENT, ERADICATION & RECOMMENDATIONS

8.1 Immediate Containment Actions (0–24 Hours):

8.1.1 [Network Containment] - Block 2 malicious external IP(s) on perimeter firewall and EDR.

8.1.2 [Endpoint Isolation] - Immediately disconnect host from corporate LAN and Wi-Fi to prevent lateral encryption spreading.

8.1.3 [Backup Verification] - Check offline immutable backups and verify integrity of VSS shadow copies on unaffected systems.

8.1.4 [Persistence Removal] - Audit and remove unauthorized Windows Registry Run keys, Scheduled Tasks, and anomalous startup services.

8.2 Short-Term Eradication (1–7 Days):

• Terminate unauthorized persistence mechanisms (Registry Run keys, scheduled tasks, and rogue services).

• Invalidate all compromised administrative credentials and force enterprise-wide Kerberos ticket resets.

• Re-image infected hosts using verified, immutable baseline golden images.

8.3 Long-Term Hardening & Post-Incident Security:

• Enforce hardware-backed Multi-Factor Authentication (MFA) across all administrative access points.

• Deploy Endpoint Detection & Response (EDR) agents with tamper protection and automated containment.

• Implement air-gapped, immutable offline backups to prevent ransomware impact.

9. CHAIN OF CUSTODY SUMMARY & CRYPTOGRAPHIC VERIFICATION

Timestamp	Action Performed	Performed By	Verification Hash / Block
2026-10-23 13:38:13	Evidence Acquisition & Cryptographic Hashing	Test Analyst	SHA256: dd0be9cf2a87b54bed5cfee78b9f...

23-Oct-2026 13:41:24 UTC+05:30	AI Forensic Triage & Multi-Parser Extraction	Cyber Triage AI Core	**Hash Chain: VERIFIED INTACT**

10. LIMITATIONS AND ASSUMPTIONS

1. **Evidence Completeness:** Findings are strictly bounded by the digital artifacts provided for triage. Unacquired physical RAM or unallocated disk sectors may harbor auxiliary attacker footprints.
2. **Obfuscation & Encryption:** Adversary anti-forensic techniques (e.g. encrypted payloads, memory unhooking) may limit static string detection.
3. **Legal Admissibility:** Cryptographic hashes and custody logs have been maintained in compliance with Section 65B of the Indian Evidence Act.

11. AUTHORISATION & FORMAL SIGN-OFF

The undersigned forensic examiners and reviewing authorities hereby certify that this investigation was conducted under rigorous forensic protocols, and the findings presented herein represent an accurate, tamper-proof analysis of the evidence.

Role	Name & Academic / Professional Designation	Signature & Official Stamp	Date
Lead Forensic Examiners	**Pragati Kiran Patil (23101024)** **Tanisha Prakash Lohar (23101009)** **Gaurav Subhash Rajguru (23101004)** B.Tech CSE (IoT, Cyber Security & Blockchain)	_____	23-10-2026
Reviewing Officer & Guide	**Ms. Tejas P. Shinde** Guide, Dept. of CSE (IoT, CS & BT), ADCET Ashta	_____	23-10-2026
Department Coordinator	**Mr. Samish N. Kamble** Project Coordinator, CSE [IOT-CSBT], ADCET Ashta	_____	23-10-2026
Head of Department	**Dr. Tahseen A. Mulla** Head of Department, CSE [IOT-CSBT], ADCET Ashta	_____	23-10-2026